Random Walks

ADVANCED TOPICS IN NETWORK SCIENCE · MODULE 07

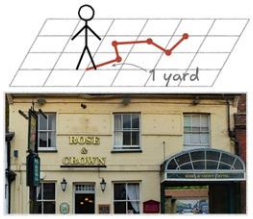
Sadamori Kojaku · Binghamton University

#1 title

Where is the walker after 100 steps?

A man walks 1 yard, turns in a random direction, and walks 1 yard again. He repeats this 100 times.

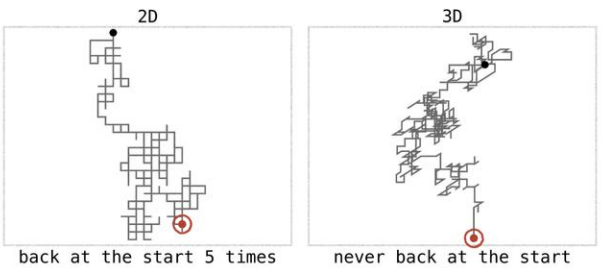
Pearson asked this in *Nature* in 1905.



Far from the start, or near it? Tell your neighbour.

Pub: Jim Osley, CC BY-SA 2.0 (geograph.org.uk)

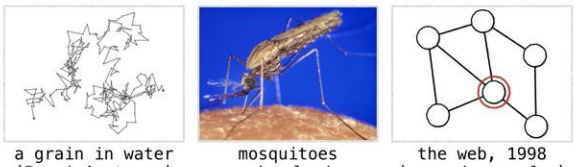
#2 hook (story): Pearson's question, 1905



2D: the walker always comes home. 3D: only 34% of walkers ever do.

#3 story: Polya's recurrence (2D returns, 3D does not)

Random walks appear in many fields



a grain in water (Perrin's trace) · mosquitoes and malaria · the web, 1998 (a random surfer)

Mosquito: CDC / James D. Gathany (public domain) · trajectory: Jean Perrin (public domain)

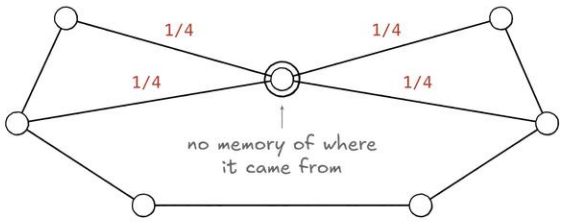
#4 hook: the same walk in three fields

How does it tell us about a network?

But it is a "random" walk, right?

#5 spine 1: the question

What is a random walk?



#6 mechanism: the rule

Pen & Paper

10 minutes · stop at the line on the sheet

#7 Pen & Paper: first block

Random walk simulator

skojaku.github.io/
adv-net-sci/assets/vis/
random-walks/index.html

#8 the simulator (QR code and link)

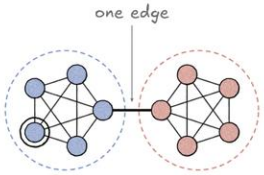
Two communities joined by one edge

How many steps, on average, until the walker crosses to the other community?

A: about 3

B: about 20

C: about 200



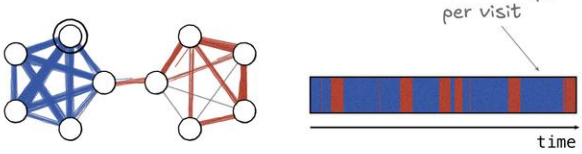
Take a guess.

9

#9 question to the room (Q2)

OPTIONAL

A short walk stays in its community



10

#10 answer to Q2

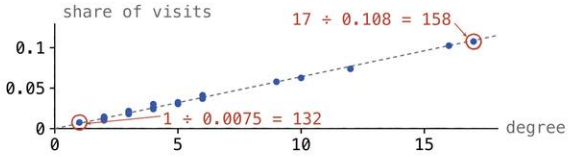
Short walks show communities. Long walks show degree. Why?

11

#11 spine 2: the mystery

Degree ÷ share is constant

Karate club: 34 nodes, 78 edges, so 2m = 156



Share of time = degree ÷ 2m

12

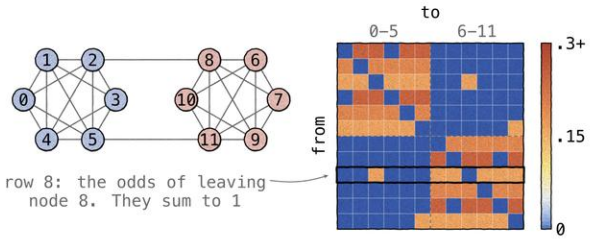
#12 debrief Part 1 (degree ÷ share)

Why does a long walk not depend on the start?

13

#13 spine 3: part question

The walk is defined by one matrix

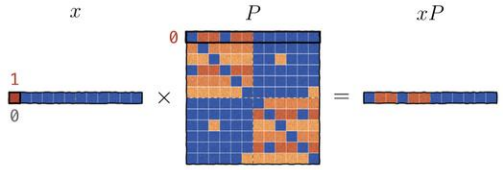


14

#14 P (the two-community network, the motif of the rest

One step is a multiplication by P

$$x(t+1) = x(t)\,P$$



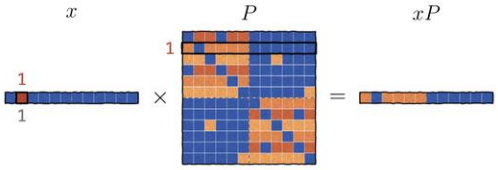
$x = (1, 0, 0, \ldots)$: xP is the row of node 0

15

#15 x P, how to read it (1/3)

One step is a multiplication by P

$$x(t+1) = x(t)\,P$$



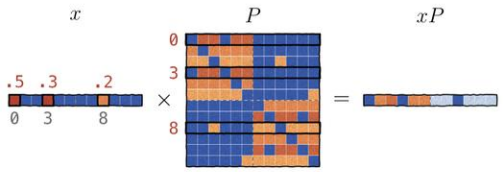
$x = (0, 1, 0, \ldots)$: xP is the row of node 1

16

#16 x P, how to read it (2/3)

One step is a multiplication by P

$x(t+1) = x(t)\,P$



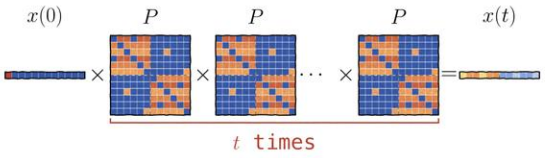
$x = (0.5, 0.05, \ldots)$: xP is a mix of the rows

17

#17 x P, how to read it (3/3)

After t steps

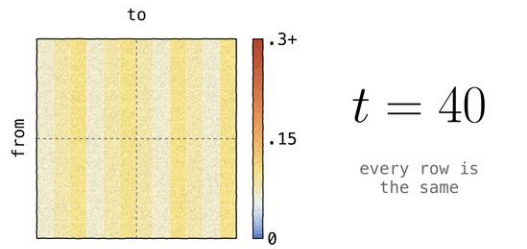
$x(t) = x(0)\,P^t$



18

#18 x(0) P^t

Rows of P^t after t steps



$t = 40$

every row is the same

19

#19 P^t as a heat map (slider)

Rows of P^t at $t = 40$

	0	1	2	3	4	5	6	7	8	9	10	11
start 0	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 1	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 2	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 3	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 4	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 5	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 6	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 7	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 8	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 9	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 10	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10
start 11	.07	.08	.10	.07	.08	.10	.08	.07	.10	.08	.07	.10

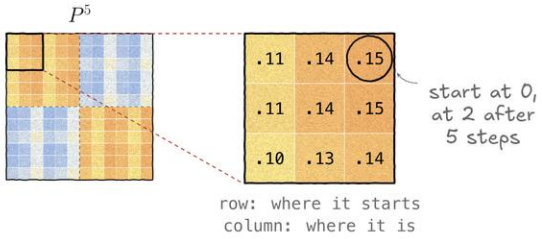
π .3+ .15 0

Every row becomes the same: π

20

#20 P^t, the numbers at t = 40

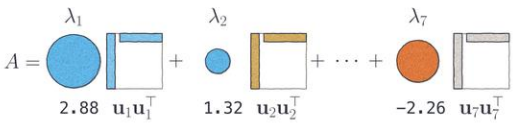
What each entry of P^t says



21

#21 zoom into P^t

Recall: eigendecomposition

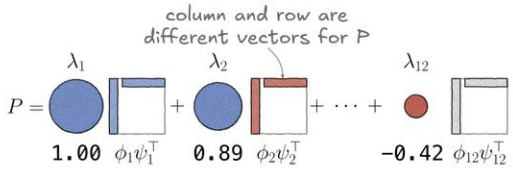


22

#22 recap: eigendecomposition (M06, light touch)

P has an eigendecomposition too

$P = \sum_k \lambda_k\,\phi_k\,\psi_k^\top$

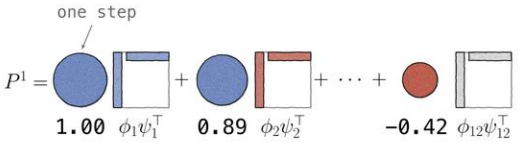


P^t changes only the eigenvalues

23

#23 P as a sum of circles and bars

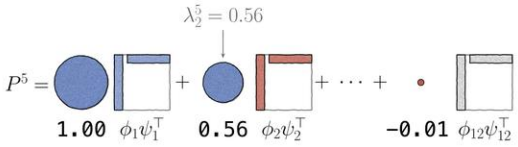
The eigenvalues shrink as t grows



24

#24 P^t as circles (1/3)

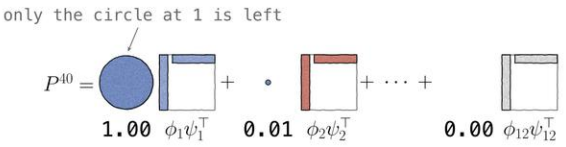
The eigenvalues shrink as t grows



25

#25 P^t as circles (2/3)

The eigenvalues shrink as t grows



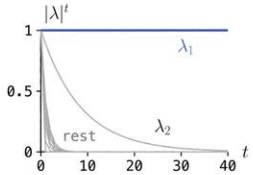
26

#26 P^t as circles (3/3)

Which term stays the longest?

The 12 eigenvalues of P.

After many steps, which term is still there?



Take 20 seconds.

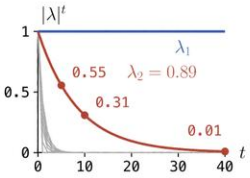
27

#27 Q3 (which term stays the longest)

The second term stays the longest

$\lambda_2 = 0.89$

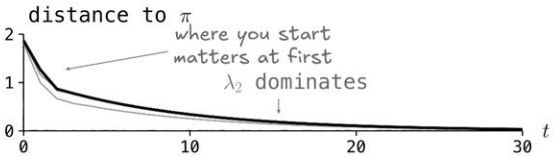
0.89^t: 0.56 at $t = 5$, 0.31 at $t = 10$, 0.01 at $t = 40$



28

#28 A3

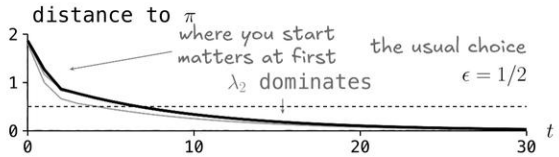
Mixing time



29

#29 mixing time (1/3)

Mixing time

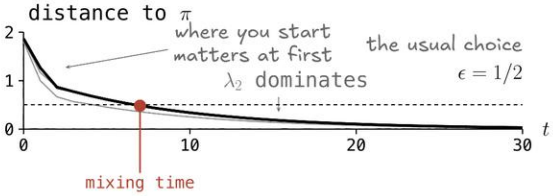


30

#30 mixing time (2/3)

Mixing time

$t_{\text{mix}} = \min\{ t : \max_{x(0)} \|x(t) - \pi\|_1 \le \epsilon \}$



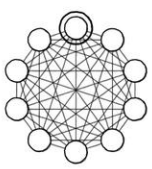
31

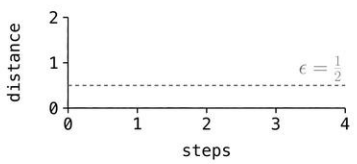
#31 mixing time (3/3)

OPTIONAL

Q: mixing time of a 10-clique?

10 nodes, every pair connected (45 edges)





Take a guess: 1 step, 5 steps, 50 steps.

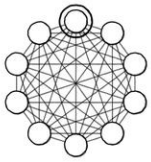
32

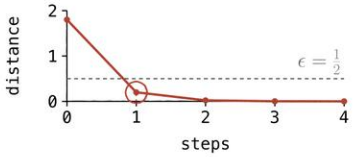
#32 Q4 (clique)

OPTIONAL

A: about 1 step

10 nodes, every pair connected (45 edges)





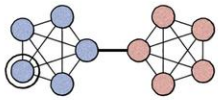
$\lambda_2 = -1/9$: **mixing takes about 1 step**

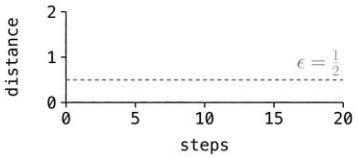
33

#33 A4

Q: mixing time with one bridge?

Two 5-cliques joined by one edge (21 edges)





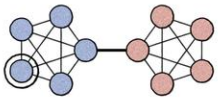
The same, 3 times longer, or 10 times longer?

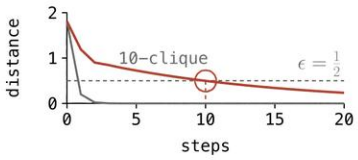
34

#34 Q5 (two cliques)

A: about 10 times longer

Two 5-cliques joined by one edge (21 edges)





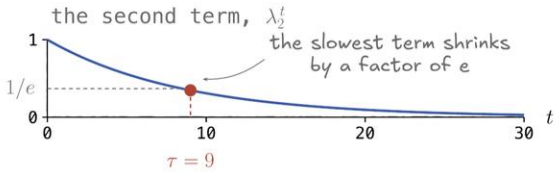
$\lambda_2 = 0.93$: **mixing takes about 10 steps**

35

#35 A5

The relaxation time

$$\tau = \frac{1}{1-\lambda_2}$$



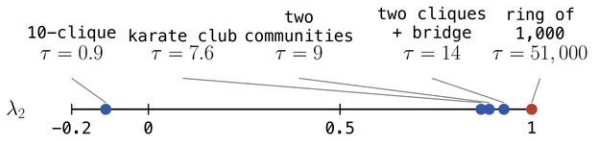
The slowest term shrinks by a factor of e in τ steps

35

#36 relaxation time

Small gap, long relaxation time

$\tau = \frac{1}{1-\lambda_2}$, the gap is $1 - \lambda_2$



The closer λ_2 is to 1, the longer τ

37

#37 spectral gap (gauge)

Mixing time is a few τ

A smaller ϵ costs a few more τ

	relaxation time τ	mixing time $\epsilon = 1/2$	mixing time $\epsilon = 0.05$
10-clique	0.9	1	2
karate club	7.6	10	27
two communities	9.0	7	27
two cliques + bridge	13.8	10	41

$$t_{\text{mix}} < \tau \log\left(1/(\epsilon\,\pi_{\min})\right)$$

38

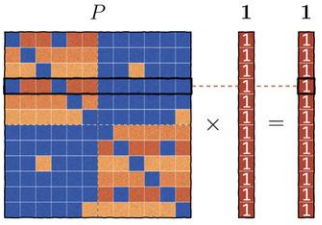
#38 mixing time versus relaxation time

One eigenvalue is always 1. Others diminish.

39

#39 spine 4: the fact

Why 1? Every row of P sums to 1



$P\,\mathbf{x} = \mathbf{x}$

↑ a vector that P does not change

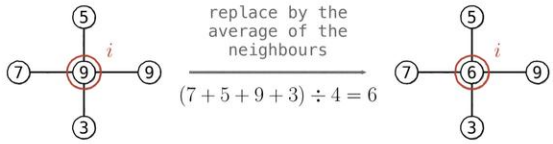
$P\,\mathbf{1} = \mathbf{1}$

40

#40 lambda = 1 exists (P 1 = 1)

No eigenvalue is larger than 1

Pf gives each node the average of its neighbours' values



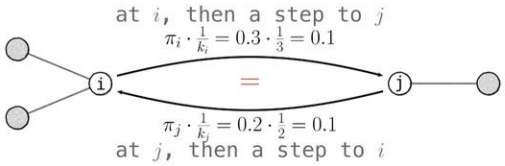
The largest value can only shrink, so $\lambda \leq 1$

41

#41 no eigenvalue above 1

Detailed balance condition

$\Pr(i \to j) = \Pr(j \to i)$



$\pi_i = k_i/2m$: proportional to degree

42

#42 the eigenvalue 1 gives pi (ties to the simulator's

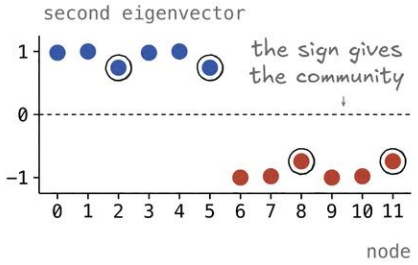
Pen & Paper

10 minutes · plot the second eigenvector

43

#43 Pen & Paper: second block

$x(t) = \pi +$ decaying terms



The slowest decaying term is the second eigenvector.

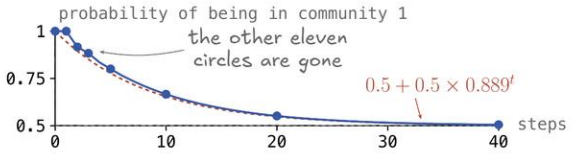
Positive: one community. Negative: the other.

44

#44 debrief Part 3 (second eigenvector; the expansion)

Early on, the community signal dominates

Walker starts at node 0

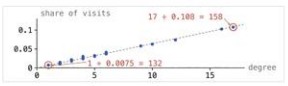


The community signal decays like λ_2^t. Degree remains.

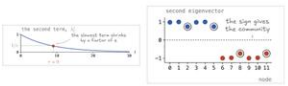
45

#45 debrief Part 3 (what dominates early)

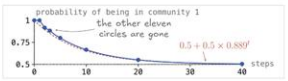
1 The eigenvalue 1 gives the degree



2 The second eigenvalue gives the communities and the mixing time



3 Short walks see communities. Long walks see degree.



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#46 summary (claims with thumbnails)